

Security Services

Security services provide protection for information systems

The major security services are

- Confidentiality
- Integrity
- Availability

These three principles form the CIA Triad

CIA Triad

- Confidentiality
- Integrity
- Availability
- Foundation of information security

CIA Triad Diagram

- Confidentiality
- Integrity
- Availability
- These three principles work together to secure information systems

Confidentiality

- Confidentiality ensures information is accessible only to authorized individuals
- Protects sensitive and private information from unauthorized access

Confidentiality



Methods to Achieve Confidentiality

- Encryption
- Access Control
- Authentication
- Password Protection
- Multi Factor Authentication

Confidentiality Example

- Only bank account holders can access their account information
- Unauthorized users are prevented from viewing sensitive data

Integrity

- Integrity ensures information remains accurate, complete and unchanged
- Unauthorized modifications are prevented

Successful Integrity



Methods to Achieve Integrity

- Hash Functions
- Digital Signatures
- Access Controls
- Checksums
- Audit Logs

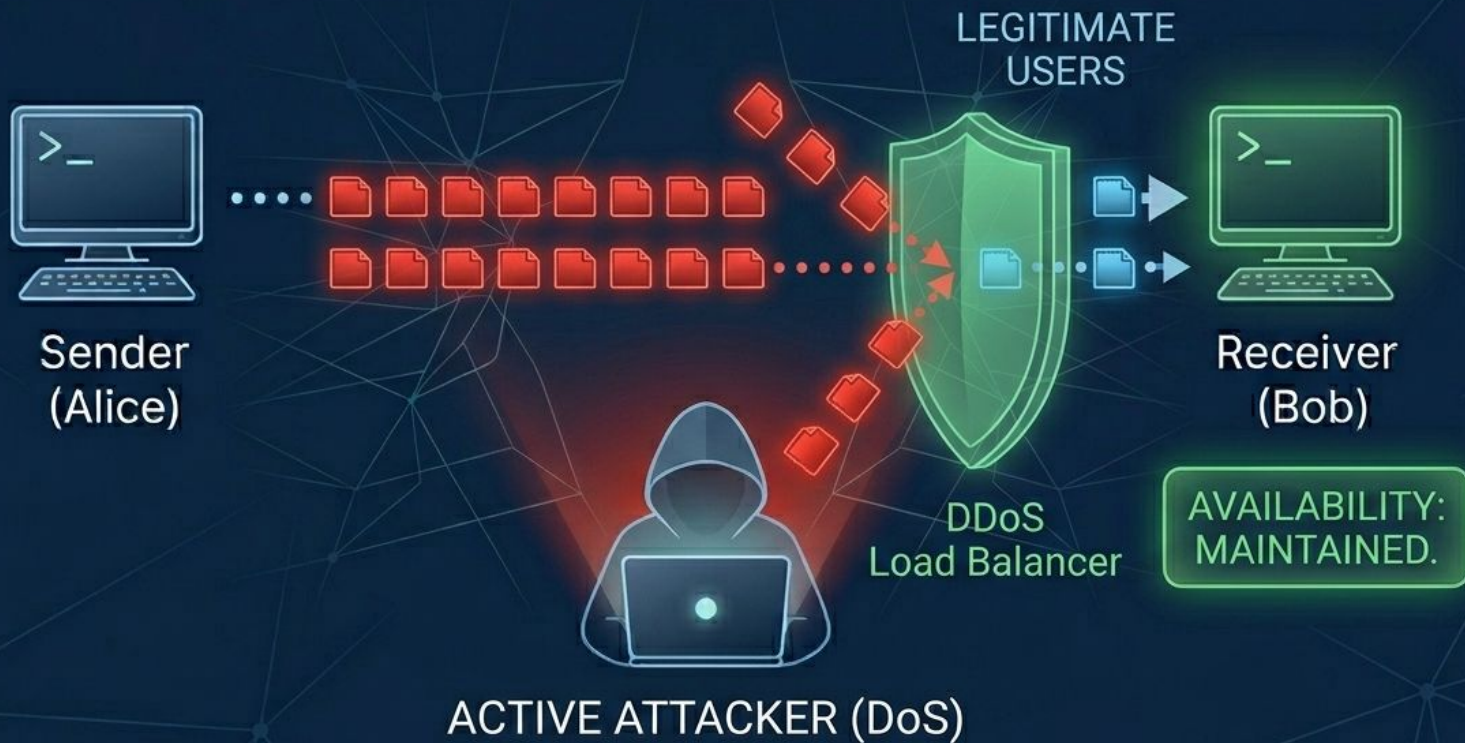
Integrity Example

- A student's marks should not be modified without authorization
- Any unauthorized change must be detected

Availability

- Availability ensures information and services are accessible whenever required
- Users can access resources without interruption

Achinst Availability



Methods to Achieve Availability

- Backup Systems
- Redundant Servers
- Disaster Recovery
- Fault Tolerance
- Regular Maintenance

Availability Example

- Online banking services should be available twenty four hours a day
- System downtime should be minimized

CIA Triad in Online Banking

- Confidentiality: Only account holders can view account details
- Integrity: Transaction data cannot be altered
- Availability: Banking services remain continuously accessible

Importance of Security Services

- Protect sensitive information
- Maintain trust in information systems
- Prevent unauthorized access and data loss
- Support business continuity

Summary

- Confidentiality protects information from unauthorized access
- Integrity ensures information remains accurate
- Availability ensures continuous access to services
- CIA Triad is the foundation of information security

Thank you

Any Questions?